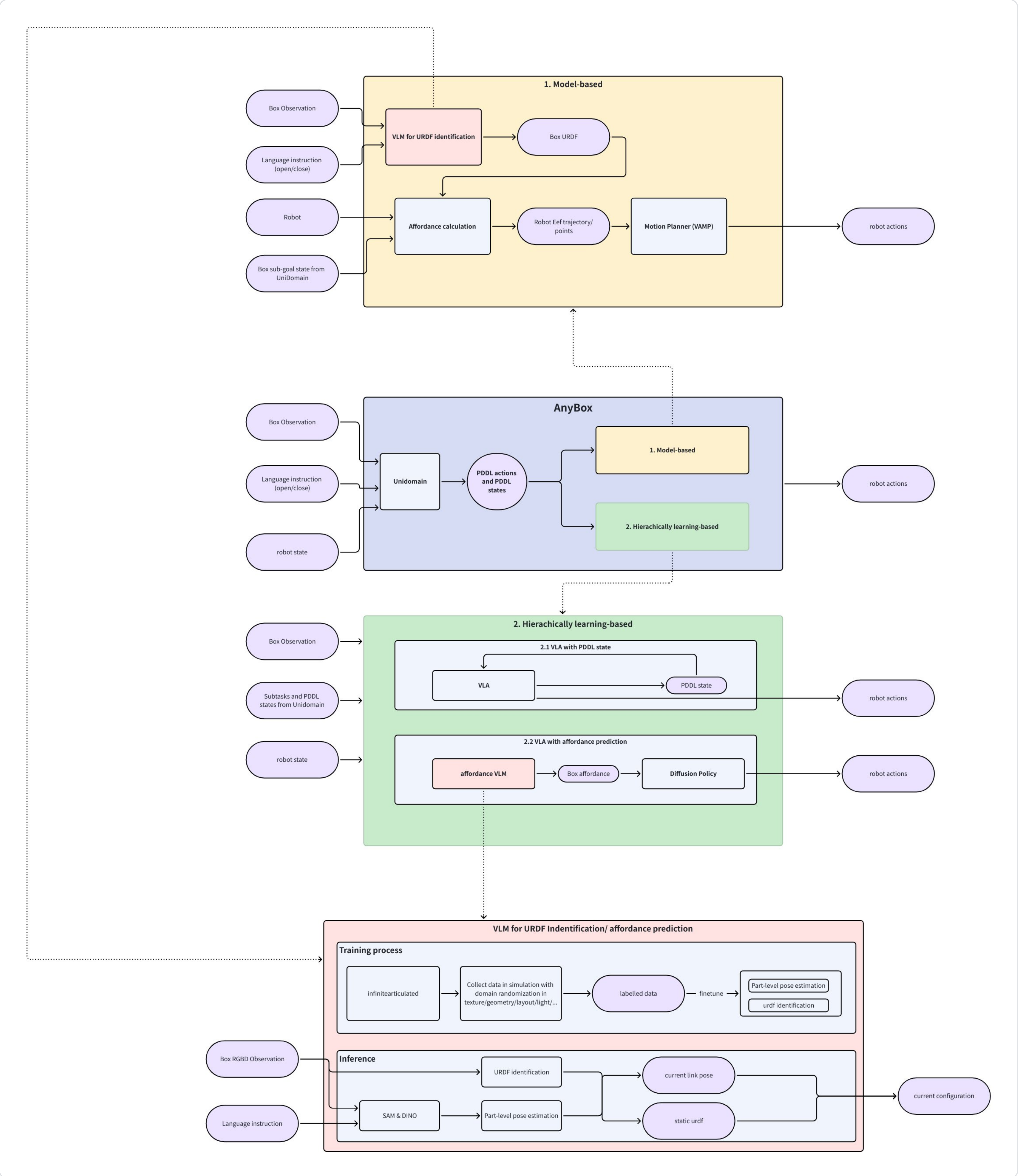
[2025.12.09] Updates

进展

Overview



VAMP

Spherization

i. 部分mesh用固定参数不能球形化
   1. Demo做法：用trimesh把它简化成形状相差很多的mesh
   2. 改进：default是`medial` (Medial Axis Transform)，用另一个octree可以。改了一下foam原始代码适配octree （去掉了效果很差的左小臂）

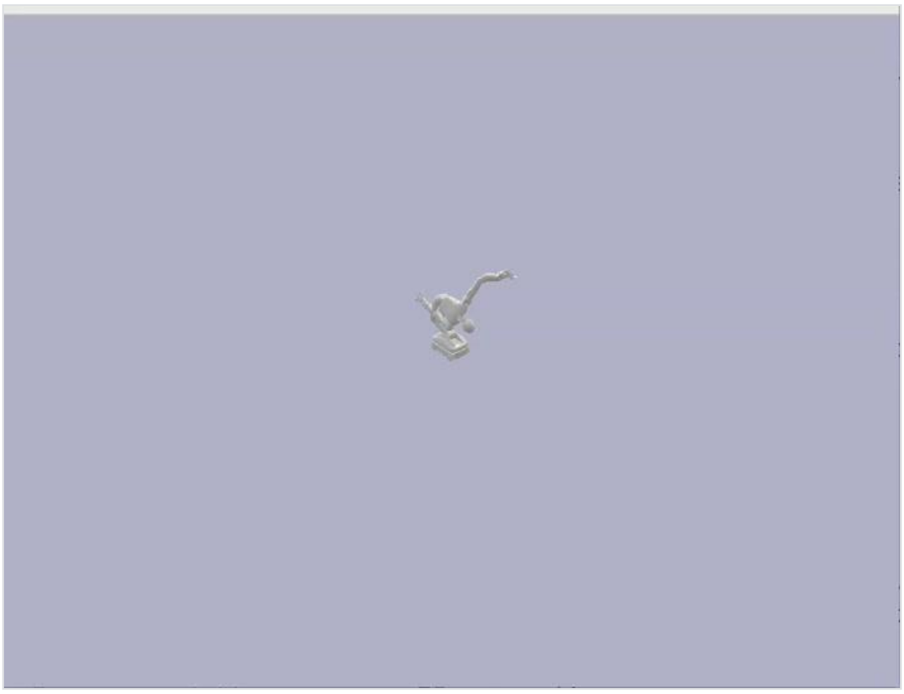
ii. 改进：去掉了一些mesh的collision，比如sensors, 还有一个部位由两个mesh组成的

Tracing compiler

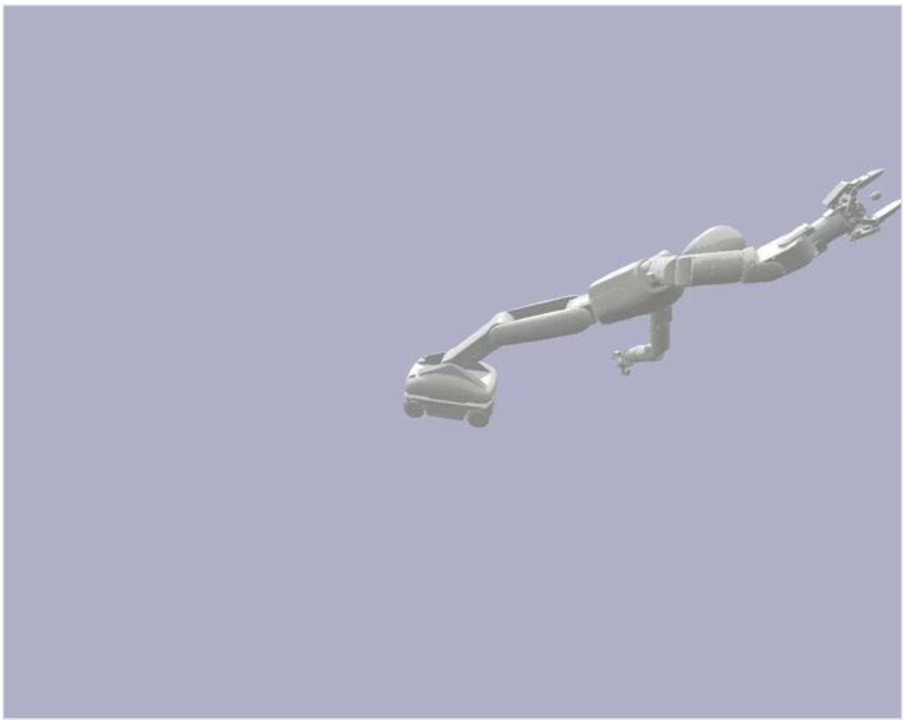
a. 之前用[zherui]的compiler，一次大概1h
b. 跑通了新开源的compiler Cricket，一次1s

Vamp

a. The sampled configuration is invalid due to self collision
   i. Demo做法：在srdf中把所有collision between link pairs 都关掉，然后运行，因为只有上半身，而且start, goal configuration差别不大，看起来没问题
   ii. 原因：randon sampled configuration看起来有问题
      1. Vamp 默认的halton sampler会返回带inf的configuration
         a. 已经解决，换成了xorshift sampler



修改前



修改后

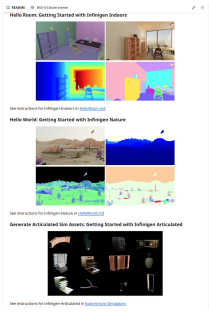
可以改进的地方

a. Extend [RAL'25] AORRTC to a Kinodynamic motion planner
b. Better spherization ([CU'25] Morphit)

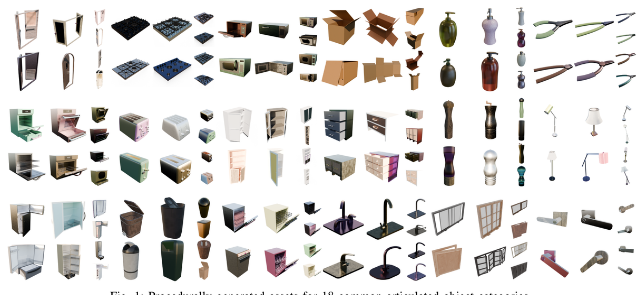
Box data

InfiniteGen Articulated from Jia Deng group, Princeton https://arxiv.org/pdf/2505.10755

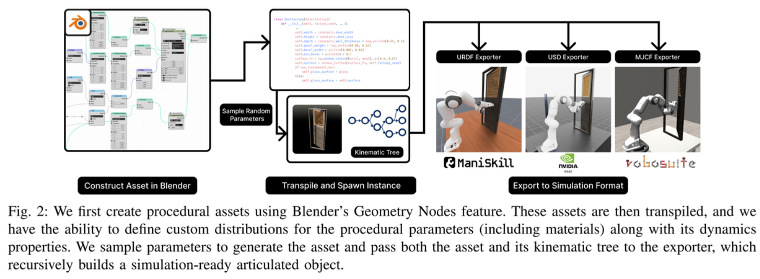
1. procedural generators for 18 common articulated object categories along with high-level utilities for use creating custom articulated assets in Blender.



InfiniteGen Nature/indoor/articulated

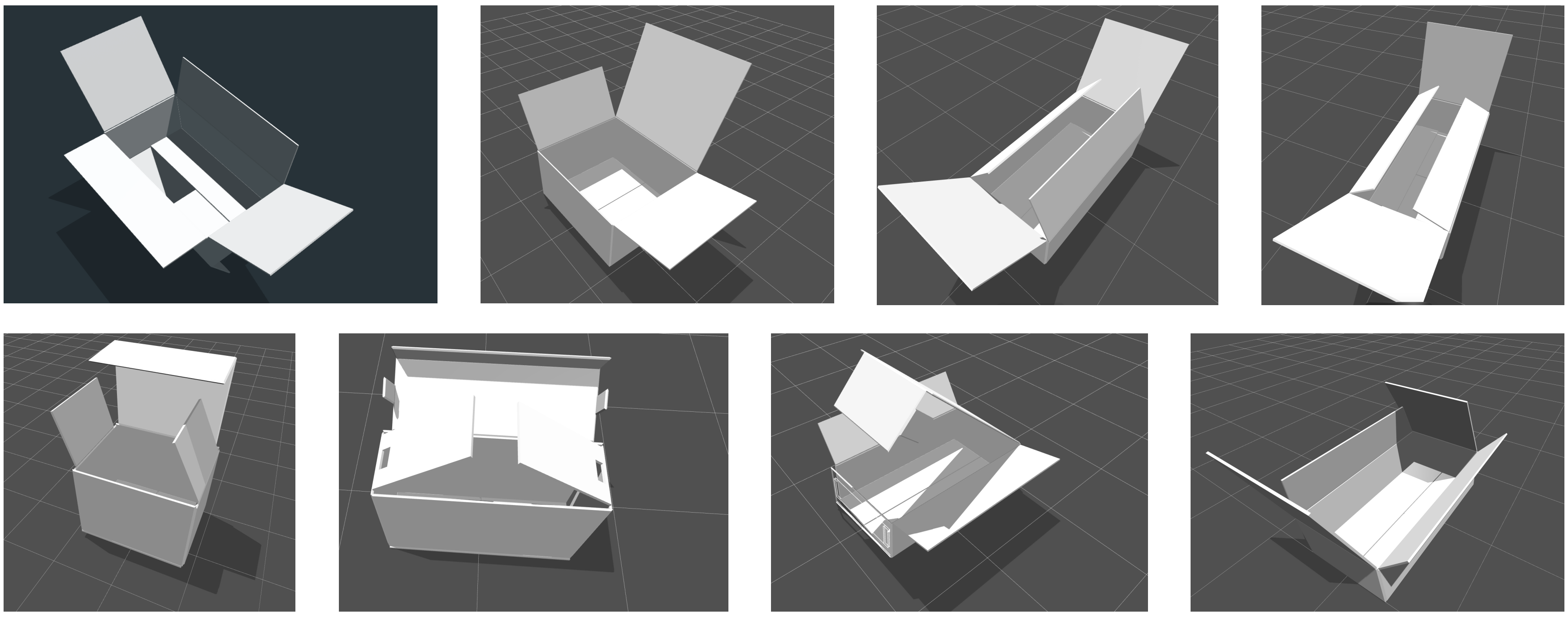


18 common articulated object categories



Overview of infinitegen articulated

   a. 8 different box types, with variants in
      i. Length, width
      ii. Texture



      i. 实际上就是三种：平口盒，插销盒和翻盖盒
      ii. 可以在源代码上加新的盒型，但是开发比较困难
      iii. Part-level primitive shape只有长方形

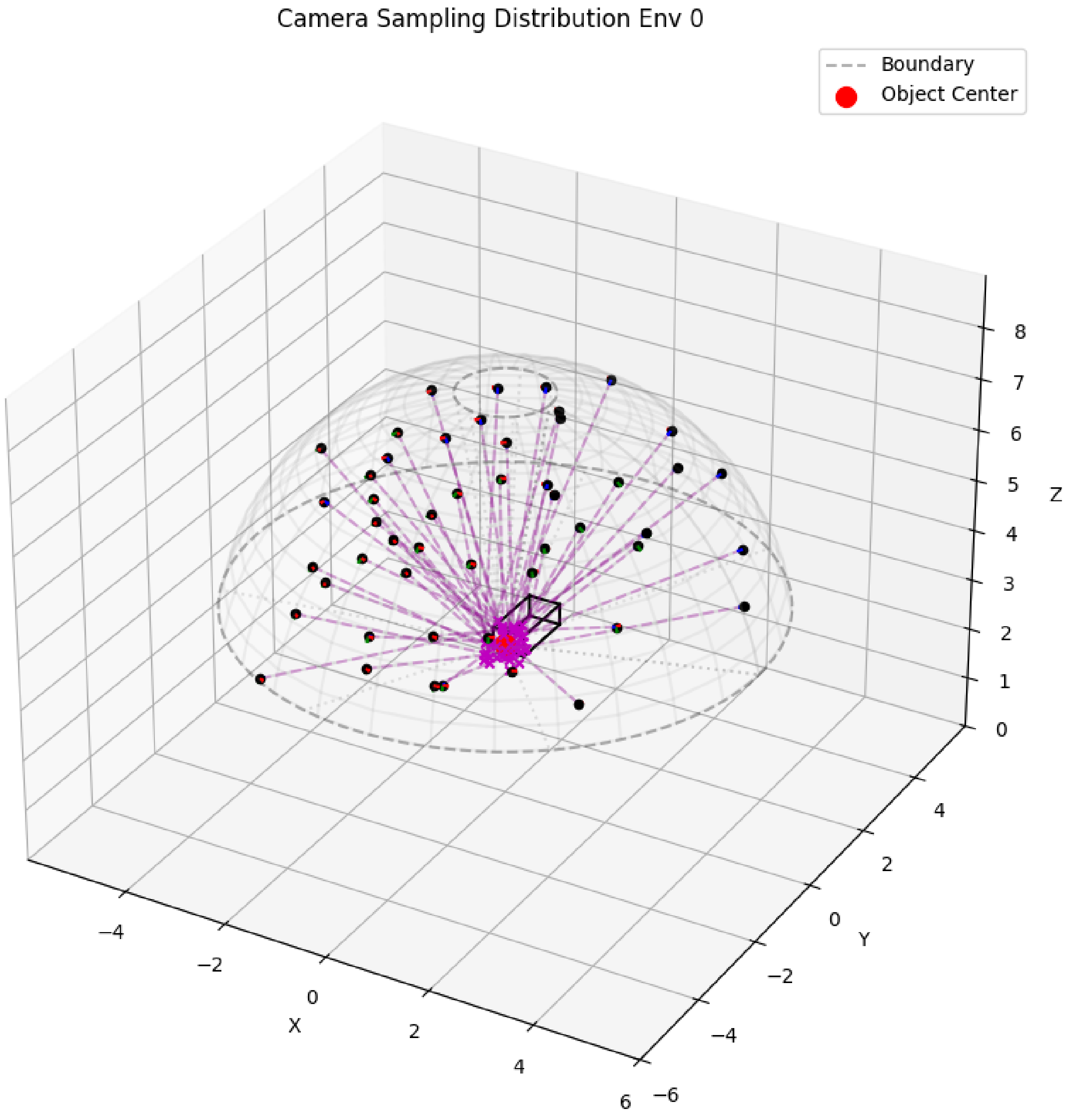
Box Data Generation

1. 采样逻辑

目前的采样逻辑可以概括为：

在以物体为中心的**球形空间内随机采样**相机位置，并对注视点施加微小抖动。生成的位姿需通过**双重筛选**：

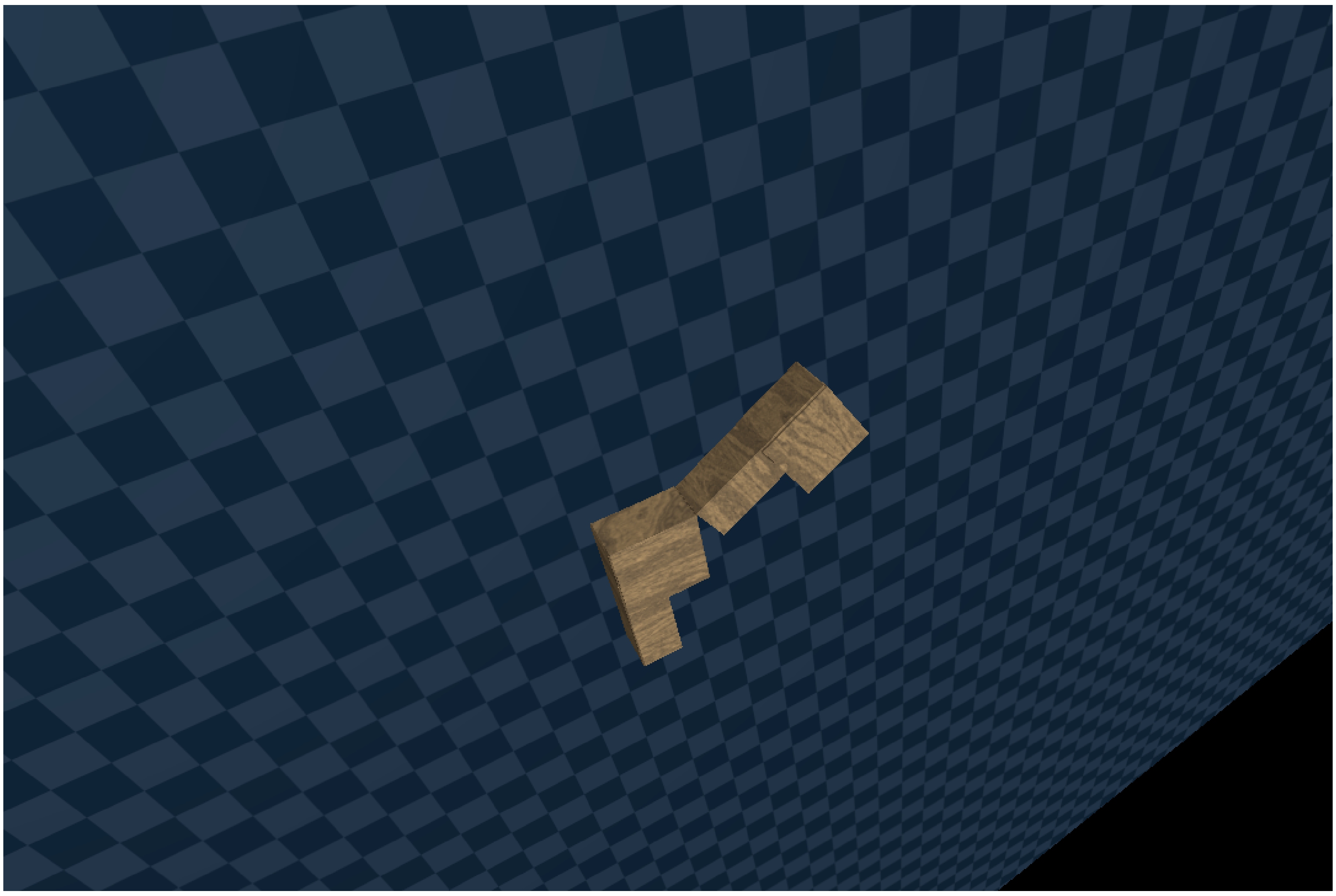
1. **可见性筛选**：确保物体的包围盒（AABB）所有角点均在相机视野内（即物体完整可见）。
2. **多样性筛选**：确保新采样点与已有采样点保持一定距离（避免视角重复）。



2. 保存内容:

   图像:

   i. Rgb



ii. Depth

iii. Segmentation (H, W), 每个value是link的index

iv. Debug video

保存了场景配置、相机参数以及每一帧渲染时的物体状态信息，所有的都在世界坐标系下.具体包括：

a. **静态/配置信息**:
   - name_idx_map: 场景中物体 Link ID 到名称的映射表。
   - `cam_intrinsics`：相机的内参矩阵。
   - `cam_pos,` `cam_lookat`, `cam_transform`：每一帧相机的世界坐标位置、观察点坐标和 4x4 变换矩阵

b. **动态/状态信息 (按时间步记录,)**:
   - `obj_pos_quat_aabb`：物体的位置、四元数姿态和 AABB 包围盒。
   - `link_pos_quat_aabb`：物体各 Link的位置、四元数姿态和 AABB 包围盒
   - `joint_pos_axis`：物体各 Joint的锚点位置和旋转轴方向。

Box URDF identification

a. What we want: a good generative model (not those retrieve-based)

b. URDF Identification reading list

c. Conclusion: [ICCV'25] PhysNAP https://arxiv.org/pdf/2508.00558 An articulation Graph Diffusion model with RGBD observation as guidance ʝ.stueckler}@uni-a.de

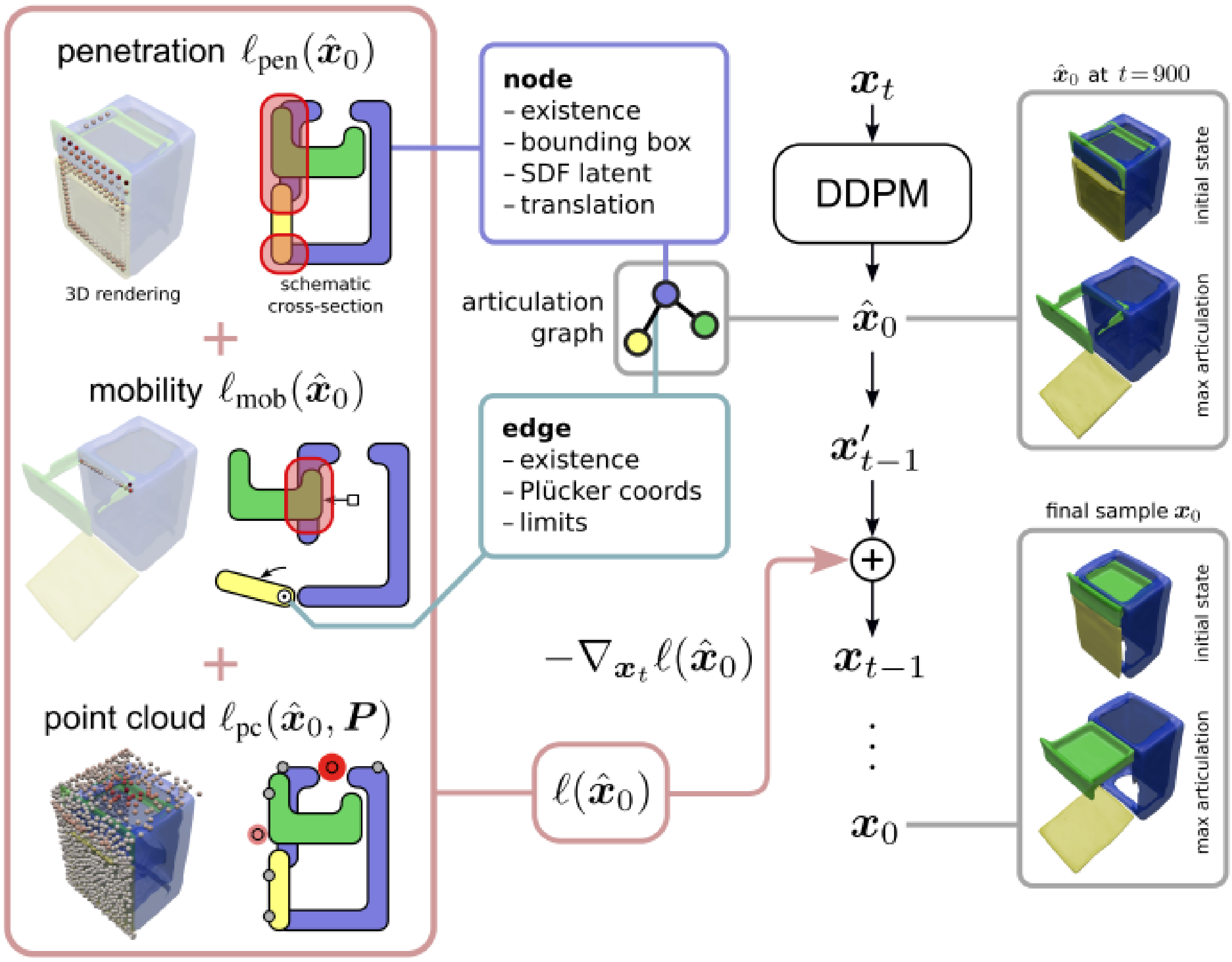


Figure 1. Method overview of PhysNAP: We propose to guide a pretrained diffusion model (NAP [10]) for articulated object generation with point cloud alignment and physical plausibility losses. The latter measure part penetration in the initial state (penetration) and in varying articulation states (mobility). All generated properties can potentially be adjusted to follow the losses, including node and edge existence, shape and articulation joint properties. We also augment NAP to condition on the object category.

i. 已经在原始数据上跑通了训练和测试代码

ii. 方案不足：

  1. 训练数据里的urdf都是拍的物体关闭状态的，observation都是从正前方拍的物体关闭状态的
  2. 所以需要能输入current configuration/结合其他工作使用

Part level pose estimation

a. [CVPR'25 highlight, Yanwei Fu] CAPNet https://arxiv.org/pdf/2504.11230v1

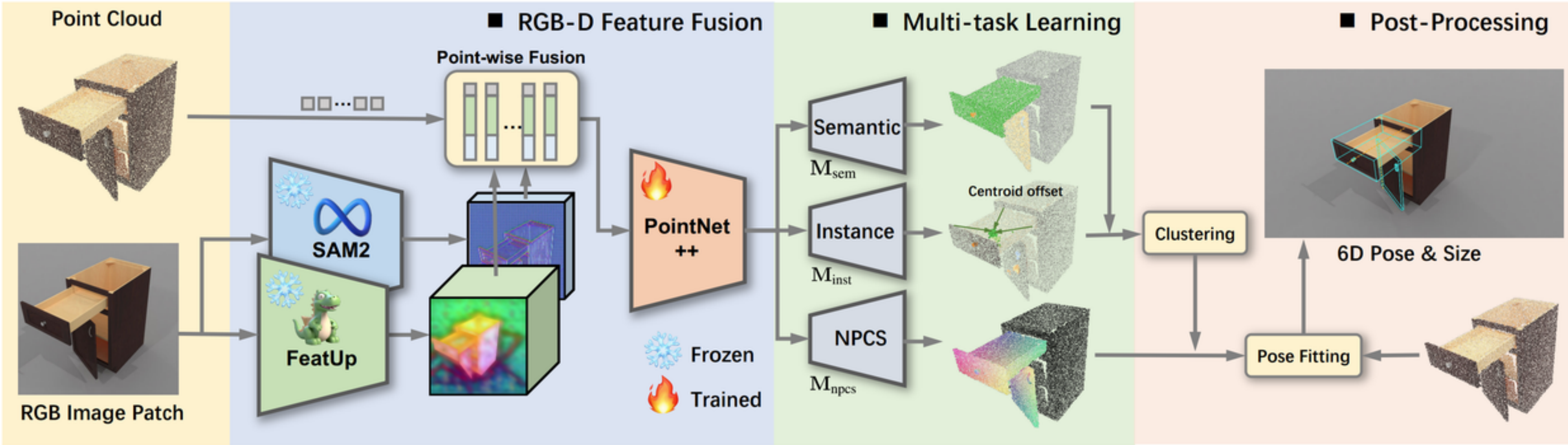
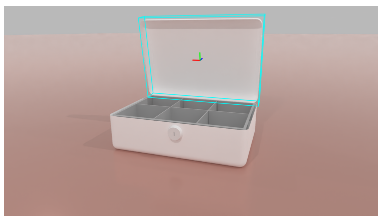
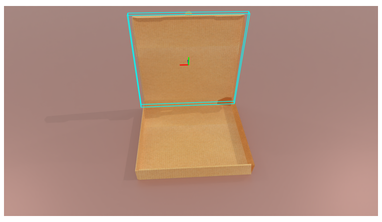
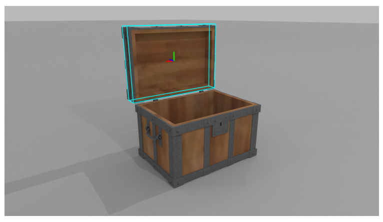
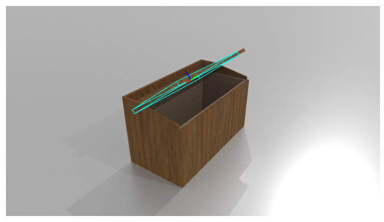


Figure 4. **Architecture overview.** CAP-Net uses pretrained vision backbones, SAM2 [28] and FeatUp [9], to extract dense semantic features, which are then fused with the point cloud in a point-wise manner. The enriched point cloud features are passed into PointNet++ [27] for further processing. These features are then used by three parallel modules to predict semantic labels, centroid offsets, and NPCS maps. A clustering algorithm groups points with the same semantic label based on centroid distances to isolate each possible part. Finally, an alignment algorithm matches the predicted NPCS map with the real point cloud to estimate each part's pose and size.

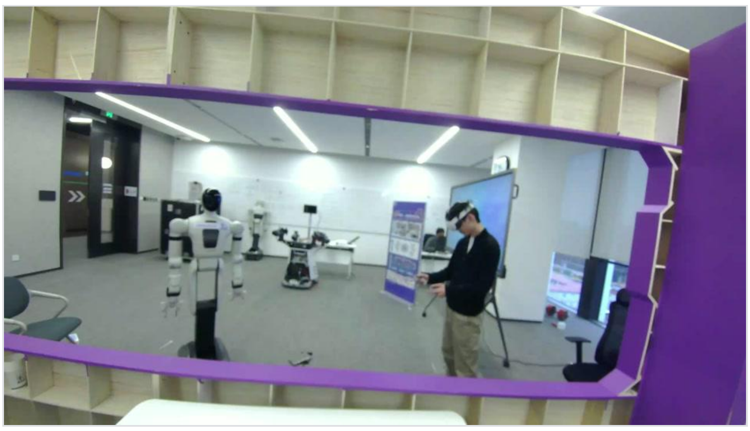
a. 在给的测试集里面跑通了评测代码，pose似乎没啥问题，但是size一般

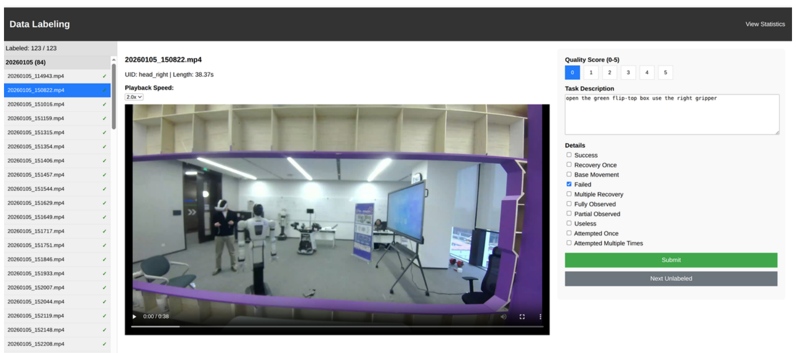
a. 和[haowu]聊了一下，后面由他继续做整个流程

Autolife manipulation data

a. 跑通了VR遥操-数据录制-打标-处理流程，预计本周可以跑通DP训练
   i. 操作盒子还是比较困难的，现在录制了120多条关盒子数据，人为进行打分标注。失败/成功/无用时间是8/24/11分钟
      1. 两条盒子数据

      2. 标注网页前端



   ii. 和[renming]聊了一下，baseline用DP和他的模型，action space是eef pose，之后可以再采一些很难的操作数据

下周计划